

Accessible Slides in LaTeX

https://github.com/rhstanton/accessible_LaTeX

Version 1.4

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Introduction

- From April 2026, course materials must meet [WCAG](#) 2.1 Level AA accessibility standards
- Applies even to password-protected course sites (e.g., [bCourses](#)).
 - (Official campus guidance issued Jan 8, 2026)
- Many of us use $\LaTeX$ to create teaching materials—both slides and documents.
- Standard $\LaTeX$ (including Beamer) does not automatically generate accessible PDFs.
- [The good news](#): The $\LaTeX$ Project team has developed solutions!

What is this project?

- An experiment exploring accessible $\LaTeX$ documents using the $\LaTeX$ Tagging Project
 - **Two templates:** Articles (`article` class) and slides (`ltx-talk`)
 - **Shows:** What is shared and what differs
 - **Contains:** Examples with math, text, graphics, and tables
 - **Scores:** Slides score 100% with [Ally](#) and pass common accessibility validators ([veraPDF](#), [PAC](#), [Acrobat](#)).

How to use it?

1. **Learn the common requirements:** Apply to any $\LaTeX$ document
 2. **See class-specific needs:** Understand slides vs. articles
 3. **Copy and adapt:** Use as templates for your documents
 4. **Study the code:** Heavily commented for learning
- Available at: https://github.com/rhstanton/accessible_LaTeX

Converting to accessible LaTeX: Overview

- **Your existing $\LaTeX$ skills transfer:** you're just adding accessibility features
- **The changes are minimal** and follow consistent patterns
- **Common requirements** (both slides & articles):
 - Add document metadata and enable PDF tagging
 - Tag images with alt text; tag table headers
 - Use accessible colors; compile with LuaLaTeX
- **Key difference:**
 - Articles: **Keep your existing class!** Just add accessibility features
 - Slides: Switch to `ltx-talk` (similar syntax, but recreate styling)
- Both require LaTeX kernel 2025-11-01 or later (see next slides for details)
- See “Getting started” slides at end for detailed steps
- [LaTeX Tagging Project documentation](#)

TeX version requirements (both slides and articles)

- **Minimum:** TeX Live 2023 or later with all packages updated
- **Critical:** Must have LaTeX kernel 2025-11-01 or later
 - Update via TeX Live Manager (Windows) or TeX Live Utility (Mac)
 - Or use Overleaf Labs (see next slide)
- **Will NOT work:** TeX Live 2022 or earlier

You **CAN** use Overleaf (both slides and articles)

- `ltx-talk` requires a very recent **LaTeX kernel** (2025-11-01 or later)
- This is available through Overleaf's [Labs program](#) (not standard Overleaf)
- **Using Overleaf (2 steps)**
 1. **Join Overleaf Labs:**
 - Visit <https://www.overleaf.com/labs/participate>
 - Opt in and enable "Rolling TeX Live releases"
 2. **Configure project:**
 - Set TeX Live version to "Rolling TeXLive (labs)" (bottom of list)
 - Set Compiler to **LuaLaTeX**
- **Resources:** <https://docs.overleaf.com/writing-and-editing/creating-accessible-pdfs>

Common requirements (1/3): Document Metadata

- **REQUIRED** for both slides AND articles
- Every accessible document **must** begin with \DocumentMetadata
- Goes *before* \documentclass

```
\DocumentMetadata{
  pdfstandard=a-2u,      % PDF/A-2u format
  pdfstandard=ua-1,      % Add PDF/UA-1 conformance target
  pdfversion=1.7,        % Explicit PDF version
  lang=en-US,            % Language for screen readers
  tagging=on,            % Enable PDF tagging

  tagging-setup={
    math/alt/use,         % Keep math accessibility enabled
    math/mathml/AF=false, % Do not embed MathML as PDF Associated Files
    math/tex/AF=false,    % Do not embed TeX source as Associated Files
    math/mathml/sources=  % Clear MathML source attachment list
  }
}
```

- Configures accessibility, math tagging, and validator-compatibility behavior in one place.

Common requirements (2/3): Images, Tables and Colors

- **REQUIRED** for both slides AND articles
- **Images:** All images must include alt text descriptions
 - Allows screen readers to describe visual content
 - Even decorative images need marking
 - See dedicated “Figures” slide for examples
- **Tables:** Must specify which rows are headers
 - Helps screen readers navigate table structure
 - Essential for data accessibility
 - See dedicated “Tables” slide for examples
- **Accessible colors:** WCAG 2.1 requires 4.5:1 contrast ratio

```
\colorlet{AccessibleRed}{red!80!black}  
\colorlet{AccessibleGreen}{green!40!black}  
% Standard blue is fine as-is
```

Common requirements (3/3): LuaLaTeX

- **REQUIRED for both slides AND articles**
- You **must** compile with LuaLaTeX (not pdfLaTeX or XeLaTeX)
- **Why LuaLaTeX?**
 1. **Automatic MathML:** Makes math accessible without manual work
 2. **Full tagging support:** Complete PDF accessibility features
 3. **Modern fonts:** Handles Unicode and OpenType fonts
- **How to switch:**
 - Command line: `lualatex filename.tex`
 - Most editors: Select “LuaLaTeX” from compiler menu

The basics

- Slides are put inside a `frame` environment, just like in Beamer.
- So **existing source files don't need a lot of editing.**
- Here's some gratuitous *math* for the accessibility checker.

```
\begin{frame}{The basics}
\begin{itemize}
\item Slides are put inside a \texttt{frame} environment, just like in Beamer.
\item So \textbf{\color{blue}existing source files don't need a lot of editing.}
\item Here's some gratuitous  $\mathit{math}$  for the accessibility checker.
\end{itemize}
\end{frame}
```

- **Note:** I set the fonts in this file so that numbers, percent signs, and dollar signs look the same in both math and text mode (my pet Beamer peeve...)
 - **Text:** \$1234567890%.
 - **Math:** \$1234567890%.

Figures

- Including figures is the same as in Beamer (e.g., using `\includegraphics`), but you need to provide a **text description**.

```
\includegraphics[height=.4\textheight,alt={A capybara}]{capybara.jpg}
```



Tables

- Including tables is the same as in Beamer (e.g., using the `tabular` environment).
- Also need to specify **header rows**. Use `{1}` for 1 header row, `{1,2}` for 2 rows, etc.
- **Note:** This example may still trigger PAC's Table header cell has no associated subcells warning even when the source tagging is correct.

```
\tagpdfsetup{table/header-rows={1,2,3}}  
\begin{tabular}{cccccrrrr}  
\toprule  
← 3 header rows  
\midrule  
← data rows  
\bottomrule  
\end{tabular}
```

Payment date	Caplet expiry date	DF_{pay}	Forward rate	Days to expiry	Days in accrual period	T_{expiry}	Δ	Caplet
2004/12/01	—	0.99550	0.01790	0	91	0.00000	0.25278	—
2005/03/01	2004/11/29	0.99008	0.02188	89	90	0.24384	0.25000	1,178.77
2005/06/01	2005/02/25	0.98401	0.02413	177	92	0.48493	0.25556	4,844.73
2005/09/01	2005/05/27	0.97733	0.02675	268	92	0.73425	0.25556	10,016.71

Common pitfalls

- Forgetting alt text for images
 - Every `\includegraphics` needs an `alt={...}` parameter
 - Even decorative images need alt text (use `alt={decorative}`)
- Not specifying table header rows
 - Add `\tagpdfsetup{table/header-rows={...}}` before each table
 - Use `{1}` for 1 header row, `{1,2}` for 2 header rows, etc.
- Insufficient color contrast
 - WCAG 2.1 requires 4.5:1 contrast ratio for normal text
 - Avoid light colors: yellow, cyan fail contrast requirements
 - Darken red and green: use `red!80!black`, `green!40!black`
 - Standard blue is fine and meets WCAG requirements
 - Test with a contrast checker: <https://webaim.org/resources/contrastchecker/>
- Using the wrong compiler
 - Make sure your editor is set to use `LuaLaTeX`, not `pdfLaTeX`
- Old TeX distribution
 - TeX Live 2022 or earlier won't work
 - Update packages using TeX Live Manager (Windows) or TeX Live Utility (Mac)

Getting started: Common steps

1. **Setup environment:** Use Overleaf Labs OR install/update TeX Live locally
 - Overleaf: See earlier slides for Labs setup
 - Local: Update via TeX Live Manager/Utility
2. **Get templates:** Download from https://github.com/rhstanton/accessible_LaTeX
3. **Add accessibility features:**
 - Add alt text to images
 - Add table/header-rows to tables
4. **Set compiler to LuaLaTeX**
5. **Compile:**
 - Compile with LuaLaTeX
 - PDF file now scores 100% on Ally

Getting started: What's different by document type

Articles (using article, report, book, etc.):

- Add `\DocumentMetadata` before `\documentclass`
- Include the tagging-setup block in `\DocumentMetadata` (as shown in this template)
- Switch to `fontspec` and `unicode-math` packages
- Keep your existing `documentclass`!

Slides (migrating from Beamer):

- Copy preamble from `accessible_slides.tex`
- Change `\documentclass{beamer}` to `\documentclass[frame-title-arg]{ltx-talk}`
- Remove Beamer themes/templates/colors
- Recreate styling using standard LaTeX/xcolor
- One-time preamble work—then reuse for all future talks!
- Questions? richard.stanton@berkeley.edu

Validation workflow: use multiple checkers

- No single checker catches everything.
- Accessibility usability checks and strict PDF conformance checks are not identical.
- A practical workflow is to run more than one checker and compare findings.
- **Checkers used so far**
 - Ally (built into bCourses)
 - veraPDF (open-source PDF conformance validator, <https://verapdf.org/>)
 - PAC (PDF Accessibility Checker, <https://pac.pdf-accessibility.org/>)
 - Adobe Acrobat.

Optional validator cleanup step: strip_af.py

- The metadata settings alone are enough to generate a 100% score on Ally.
- Stricter checkers (e.g., veraPDF) may still complain about /AF or embedded-file structures.
- To remove these and stop the complaints, run strip_af.py:

```
python strip_af.py build/accessible_slides.pdf
```

- Prerequisite: install the Python package pikepdf first: `python -m pip install pikepdf`.
- Then validate build/accessible_slides.pdf with veraPDF.

Validator-specific notes

- **Acrobat: “Lb1 and LBody – Failed” (validator bug)**
 - Appears around figure captions, section numbers, table captions
 - This is an Acrobat bug, not a source error – safe to ignore
 - The <Lb1> tag is correctly used for numbering
 - Acrobat’s checker incorrectly expects it only in lists
 - Reference: <https://tex.stackexchange.com/a/709655/2388>
- **PAC table warning (known behavior)**
 - PAC can report `Table header cell has no associated subcells` even when `table/header-rows` is correct
 - This behavior is documented by the LaTeX Tagging Project: <https://github.com/latex3/tagging-project/issues/1056>
- **When this warning appears**
 - First confirm source markup is correct (especially `table/header-rows`)
 - Keep the source tagging and document the validator warning
 - Report minimal reproductions to tool vendors when possible
- **Bottom line**
 - Use multiple validators (Ally, veraPDF, PAC, Acrobat) and compare findings
 - Treat one-tool warnings as signals to investigate, not proof of errors